

CPR Esoft Quiz 2

Question #: 1

A 20-year-old man comes to the physician for a routine examination. He has no significant past medical history and takes no medications. Physical examination shows eruptive xanthomas. Fasting blood samples are obtained for lipid studies. The fasting serum sample appears milky and turbid. A creamy layer is observed after centrifugation. Results of laboratory studies are shown in the table. Which of the following proteins is most likely deficient in this patient?

- A. Apolipoprotein E
- B. Apolipoprotein B100
- C. LDL receptor
- ✓D. Lipoprotein lipase (LPL)
- E. Lecithin cholesterol acyltransferase (LCAT)
- F. Cholesterol-ester transfer protein (CETP)
- G. Microsomal triglyceride transfer protein (MTP)

Rationale: The patient in the vignette has highly elevated serum triglyceride concentration. Serum triglycerides is mainly associated with chylomicrons and VLDL. Decreased activity of lipoprotein lipase (Or Apo C deficiency) results in a defect in the breakdown of triglycerides in chylomicrons, resulting in very high serum triglyceride concentration. This is also known as type-I hyperlipoproteinemia.

LPL (Or Apo C deficiency) deficiency and overproduction of VLDL may lead to high VLDL found in patients with type IV hyperlipoproteinemia, which is more common than the rare type I. Both type I and type IV have turbid samples but type I has a creamy layer on top after centrifugation and generally much higher triglyceride levels.

Review the functions of the other proteins:

Apo E deficiency results in elevated serum TAG and elevated serum cholesterol (Type III hyperlipoproteinemia).

Apo B100 mutations (present in LDL and recognized by the LDL-receptor) can result in elevated serum cholesterol levels (less common form of familial hypercholesterolemia).

LDL receptor mutations (recognize apo B100 on LDL) result in familial hypercholesterolemia.

LCAT is required for reverse cholesterol transport. It transfers fatty acid from lecithin (phosphatidylcholine) to cholesterol to form cholesterol ester. Cholesterol ester is then transferred into the core of HDL.

Cholesterol ester transfer protein is responsible for the exchange of lipids between VLDL and HDL.

Cholesterol ester is transferred from HDL to VLDL, and triglycerides are transferred from VLDL to HDL.

Microsomal TAG-transfer protein (MTP) is involved in the assimilation of triglycerides to chylomicrons in the intestinal mucosal cell and addition of triglycerides to VLDL in the liver cells. Deficiency of MTP results in abetalipoproteinemia/ hypobetalipoproteinemia, and very low serum triglyceride and serum cholesterol concentration.

Attachment:

Test	Result	Reference Range
Total cholesterol	180 mg/dL	<200 mg/dL
Triglycerides	900 mg/dL	<150 mg/dL
HDL cholesterol	40 mg/dL	>40 mg/dL
LDL cholesterol	90 mg/dL	<130 mg/dL

92786.jpg

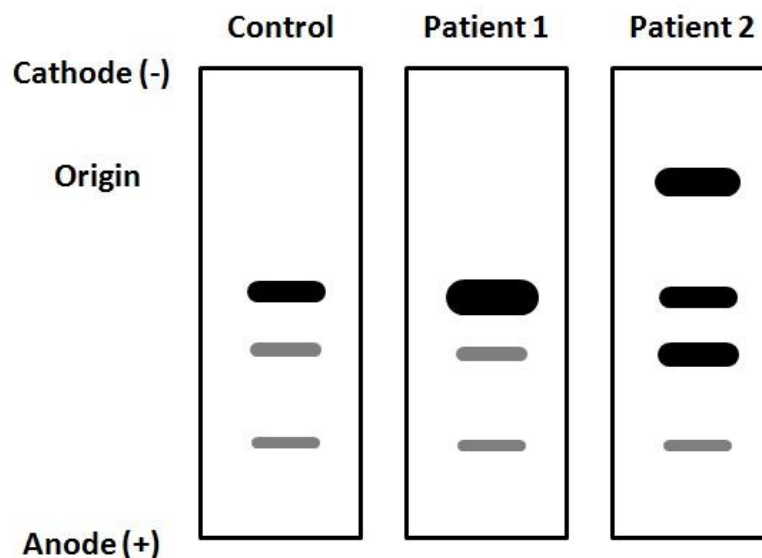
Question #: 2

Fasting serum lipoprotein electrophoresis is performed on samples from two patients with lipoprotein disorders and a normal control sample. The results of the test are shown. Which of the following is the best interpretation of these laboratory findings?

- A. Patient 1 has Type IIB hyperlipoproteinemia
- ✓B. Patient 2 has Type V hyperlipoproteinemia
- C. Patient 1 would have high serum triglyceride levels
- D. Patient 1 has a mutation in the ApoE (remnant) receptor
- E. Patient 2 would have high serum total cholesterol and high LDL cholesterol

Rationale: Patient 1 has increased staining at the beta-lipoprotein, which indicates elevated levels of LDL (Type IIA hyperlipoproteinemia). Patient 1 would also have elevated serum cholesterol levels. The most common defect is a mutation of the LDL receptor. Patient 2 has increased staining at the origin (chylomicrons) and pre-beta lipoprotein (VLDL), indicating Type V hyperlipoproteinemia. Patient 2 would also have an elevated serum triglyceride concentration.

Attachment:



64546.jpg

Question #: 3

A 40-year-old man comes to the physician because of a 2-day history of tightness in the chest while walking uphill. His father died of a myocardial infarction at the age of 55 years. His blood pressure is 160/95 mmHg. Physical examination shows palmar xanthomas. Results of fasting laboratory studies are shown in the table. Coronary angiography shows atherosclerotic narrowing of multiple coronary arteries. The patient's condition is most likely caused by a defect in which of the following processes?

- ✓A. Receptor mediated endocytosis by Apo E
- B. Synthesis of Apo B-48
- C. Mutations in the LDL receptor gene
- D. Expression of lipoprotein lipase
- E. Production of lecithin cholesterol acyltransferase

Rationale: The vignette describes a patient with familial dysbetalipoproteinemia. Note the palmar xanthomas, elevated serum triglycerides (TAG), and cholesterol concentration. It is characterized by elevated levels of IDL. The most common defect is the receptor-mediated endocytosis of remnant lipoproteins due to a mutation in the gene encoding Apo E.

Mutations in the LDL receptor gene result in familial hypercholesterolemia (Type IIA), characterized by elevated serum LDL and cholesterol levels.

Defective activity of lipoprotein lipase (or Apo C deficiency) results in type I hyperlipoproteinemia, characterized by elevated serum triacylglycerol and chylomicrons.

LPL deficiency (or Apo C deficiency) and overproduction of VLDL lead to elevated triglycerides (TAG) in Type IV hyperlipoproteinemia.

Attachment:

Test	Result	Reference Range
Total cholesterol	340 mg/dL	<200 mg/dL
Triglycerides (TAG)	300 mg/dL	<150 mg/dL
HDL cholesterol	38 mg/dL	>40 mg/dL

92788.jpg

Question #: 4

A 35-year-old man is brought to the emergency department because of a sudden onset of crushing substernal chest pain that radiates to his neck. He has had prior episodes of exertional chest pain relieved by rest. His father died at the age of 40 years from a myocardial infarction. Fasting serum lipoprotein electrophoresis shows prominent Beta and Pre-Beta bands, with a decreased alpha lipoprotein band. Which of the following is the most likely lipid disorder in this patient?

- A. Type IIA (Familial hypercholesterolemia)

- ✓B. Type IIB (Familial combined hyperlipidemia)
- C. Type III (Familial dysbetalipoproteinemia)
- D. Type I hypertriglyceridemia
- E. Type IV hypertriglyceridemia
- F. Hypoalphalipoproteinemia
- G. Hypobetalipoproteinemia

Rationale: The vignette describes a patient with a family history of coronary artery disease. Lipoprotein electrophoresis shows increased beta lipoprotein (LDL) and pre-beta lipoprotein (VLDL), and low alpha-lipoprotein (HDL). This is diagnostic of type II B hyperlipoproteinemia (Familial combined hyperlipidemia). Patients with type IIB also have increased serum cholesterol and serum triglycerides (TAG) concentration.

Type IIA is characterized by increased risk of atherosclerosis and coronary artery disease and increased levels of beta lipoprotein (LDL).

Type III is characterized by an increased staining between the beta and the pre-beta lipoprotein bands and an increase in the remnant lipoprotein (IDL).

Type I is characterized by an increased staining of the chylomicron region (origin).

Type IV is characterized by an increased staining of the pre-beta lipoprotein (VLDL).

Hypoalphalipoproteinemia is found in Tangier disease. Characterized by very low HDL cholesterol.

Hypobetalipoproteinemia leads to very low levels of chylomicrons, LDL, and VLDL. Serum total cholesterol and serum triglycerides (TAG) are low.

Question #: 5

A 3-month-old boy is brought to the physician by his mother because of recurrent vomiting, diarrhea, and poor weight gain since birth. Physical examination shows abdominal distension. Laboratory studies show increased fecal fat and steatorrhea. The results of the fasting serum lipid profile are shown in the table. He is prescribed a specialized diet that includes supplementation with vitamins A, D, E, and K. Which of the following proteins is most likely deficient in this patient?

- A. Membrane ABCA1-cholesterol transporter
- B. Apolipoprotein A-I
- C. Apolipoprotein C-II
- D. Cholesteryl ester transfer protein (CETP)
- ✓E. Microsomal triglyceride transfer protein (MTP)
- F. Lecithin cholesterol acyl transferase (LCAT)

Rationale: The patient has abetalipoproteinemia; Note the malabsorption, steatorrhea, very low serum triglycerides, and cholesterol levels.

This is due to Microsomal triglyceride transfer protein (MTP) deficiency. MTP is required in the intestinal mucosal cells for chylomicron assembly and secretion into lymphatics; It is also required for assembly and release of VLDL from the liver. MTP deficiency results in very low chylomicrons (leading to

malabsorption); and very low VLDL and LDL levels (Leading to very low serum triglycerides and cholesterol levels).

Tangier disease is due to membrane ABCA1 cholesterol transporter deficiency. Presents as orange tonsils and very low HDL-cholesterol levels.

ApoA-I deficiency leads to low HDL levels and reduced activity of lecithin cholesterol acyl transferase (LCAT). LCAT deficiency and ApoA-I deficiency can increase the risk of cardiovascular disease due to low HDL levels.

Apo-CII deficiency results in reduced activity of endothelial lipoprotein lipase. Serum triglycerides may be elevated (Type-I, Type IV, or type-V hyperlipoproteinemia).

CETP transfers Cholesterol from HDL to VLDL and triglycerides from VLDL. Increased CETP activity is associated with increased risk of atherosclerosis (due to elevated VLDL and LDL levels)

Attachment:

Test	Result	Reference Range
Triglycerides	15 mg/dL	<150 mg/dL
Total cholesterol	40 mg/dL	<200 mg/dL

311715.jpg

Question #: 6

A 60-year-old man comes to the physician for a follow-up evaluation. He has a 3-year history of type 2 diabetes mellitus and hypertension. Six months ago, he began taking his medications regularly after experiencing a myocardial infarction. He reports that since then, he has made several lifestyle changes: he reduced red meat intake, walks for 30 minutes daily, quit smoking, and attends regular follow-up visits every 2 months. His blood pressure measurements over the past several visits have ranged from 136–140/80–88 mm Hg.

Results of laboratory studies are shown:

Hemoglobin A1c: 6.5% (Reference: 5.7-6.4%)

Serum triglycerides: 180 mg/dL (Reference: <150 mg/ dL)

HDL-cholesterol: 30 mg/dl (Reference: >40 mg/ dL).

Dietary supplementation of which of the following nutrients will most likely positively affect his HDL levels?

- A. Trans fatty acids
- B. Palmitic acid
- ✓C. Oleic acid
- D. Alpha-linolenic acid
- E. Folic acid
- F. Ascorbic acid

Rationale: Oleic acid (monounsaturated fatty acid), rich in Mediterranean diets, is reported to increase HDL levels.

Trans fatty acids and palmitic acid (a saturated fatty acid) increase the risk of cardiovascular disease by

increasing LDL-cholesterol levels.

Alpha-linolenic acid is an omega-3 fatty acid that reduces the risk of cardiovascular disease but has no effect on HDL or LDL levels.

Folic acid (vitamin B9), vitamin B12, and vitamin B6 reduce serum homocysteine levels.

Question #: 7

An investigator is studying tissue-specific processing of the apolipoprotein B (ApoB) pre-mRNA. It is observed that an enzymatic modification of a single base results in different ApoB mRNAs being produced in the liver and intestinal mucosal cells. In the liver, the full-length ApoB-100 protein is synthesized, whereas in the intestinal mucosal cells, a shorter ApoB-48 protein is produced from the same gene. Which of the following mechanisms is most likely responsible for producing this diversity of proteins in these cells?

- ✓A. RNA editing
- B. Alternative splicing
- C. Poly-adenylation
- D. Proteolytic cleavage
- E. RNA capping

Rationale: The Apo B48 mRNA in the intestinal mucosal cell is synthesized as a large precursor.

Post-transcriptional editing (RNA editing) of a codon changes an amino acid codon to a STOP codon.

Hence, the protein synthesized in the intestinal mucosal cell is shorter than Apo B100 synthesized in the liver.

Note: Apo B48 is a component of chylomicrons, and Apo B100 is a component of VLDL, LDL, and IDL.

Question #: 8

A 56-year-old man is brought to the emergency room because of a 2-hour history of shortness of breath, nausea, and palpitations. His pulse is 132/min, respirations are 18/min, and his blood pressure is 162/96 mmHg. Cardiac auscultation shows an irregular heartbeat, varying from 120-140 bpm. ECG confirms atrial fibrillation. Administration of an experimental drug that inhibits the T-type calcium channels, lowers his heart rate to 92/min. There is no change in cardiac contractility or diastolic aortic pressure. This drug is most likely to cause which of the following effects on the cardiac action potentials?

- ✓A. A
- B. B
- C. C
- D. D

E. E

Rationale: Phase 4 is the spontaneous depolarization (pacemaker potential) that triggers the action potential once the membrane potential reaches threshold between -40 and -30 mV). The T-type voltage gated calcium channel are important in generating potentials that allow for rhythmic bursts of action potentials in cardiac cells of the sinoatrial node of the heart. A T-Type calcium channel blocker would reduce the depolarizing activity of Phase 4 increasing the refractory period.

Attachment:

	SA node affected phase	SA node depolarization rate
A	Phase 4	Decreased
B	Phase 0	Increased
C	Phase 4	Increased
D	Phase 0	Decreased
E	Phase 3	Increased

Capture.PNG

Question #: 9

An 83-year-old man comes to the physician because of a 1-week history of shortness of breath. He has a 20-year history of hypertension and a 10-year history of coronary artery disease. His pulse is 100/min, blood pressure is 164/94 mm Hg, and respirations are 20/min. Physical examination shows elevated jugular venous pulse, pitting edema of the ankles and bilateral basal lung crepitations. He is diagnosed with congestive heart failure. Cardiac catheterization study shows:

Systemic arterial oxygen content =0.2 mL O₂/mL,

Pulmonary arterial oxygen content =0.1 mL O₂/mL and

Total oxygen consumption =400 mL O₂/min.

Using the above data, which of the following values best represents this patient's stroke volume?

- A. 30 mL/beat
- ✓B. 40 mL/beat
- C. 50 mL/beat
- D. 60 mL/beat
- E. 70 mL/beat

Rationale: First find out cardiac output from the given data:

CO = O₂ consumed/Arteriovenous diff of O₂

=400/(0.2-0.1) = 4L/min.

Next, find out SV:

SV = CO/Heart rate

=4000/100 = 40 mL per beat.

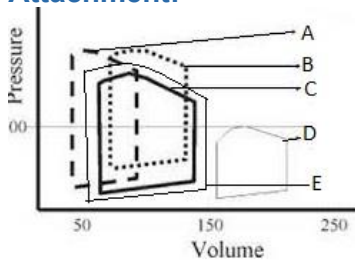
Question #: 10

A 62-year-old woman comes to the physician because of weakness, fatigue, fever, and weight gain 5 years following her radiation therapy post breast cancer surgery. She also complains about abdominal discomfort and difficulty breathing doing work. Physical examination shows hepatomegaly and jugular venous distention that fails to subside on inspiration but shows no evidence of hypotension or pulsus paradoxus. An echocardiogram shows reduced end-diastolic volumes, elevated diastolic pressures in both ventricles and a slightly elevated end-systolic volume. Which of the following pressure-volume loops will most likely be associated with this patient?

- A. A
- ✓B. B
- C. C
- D. D
- E. E

Rationale: reduced end-diastolic volumes, elevated diastolic pressures in both ventricles and a slightly elevated end-systolic volume

Attachment:



PV LOOPS.jpg

Question #: 11

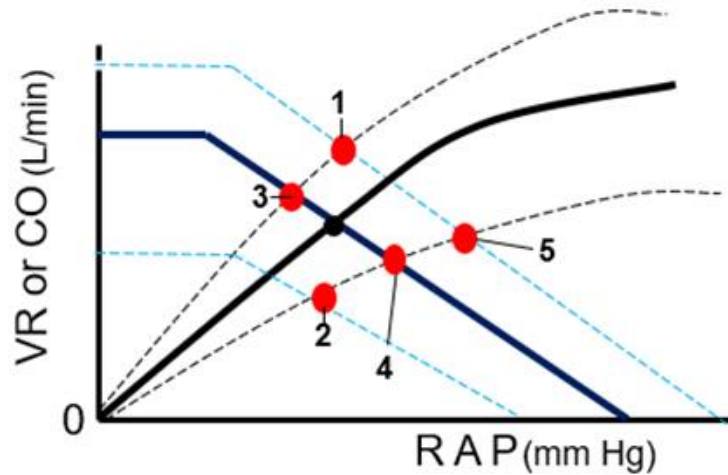
A 65-year-old woman is brought to the emergency department because of worsening shortness of breath and swelling in both feet. She has a 3-month history of decompensated heart failure. Which of the labeled points on the attached Guyton cross plot most likely reflects compensated heart failure?

- A. 1
- B. 2
- C. 3
- D. 4
- ✓E. 5

Rationale: In decompensated heart failure, contractility decreases, cardiac output decreases, and the cardiac performance curve is shifted to the right and downwards as it represented by the number 4. When CO decreases then venous return also decreases and the body tries to compensate. This is

termed compensated heart failure when in an effort to increase VR and CO the sympathetic nervous system causes increased venoconstriction and increased blood volume. This affects the vascular function curve shifting it upwards and to the right as seen in number 5.

Attachment:



CrossPlot.png

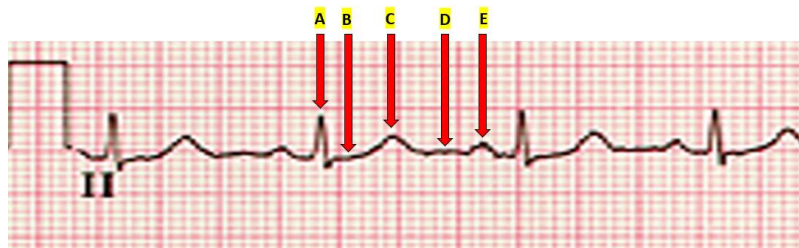
Question #: 12

A 35-year-old man comes to the physician for a routine examination. He has been part of the national cycling team for 15 years. His blood pressure is 126/76 mmHg, pulse is 61/min and respirations are 12/min. Cardiac examination shows a sustained apical impulse that is laterally displaced. Echocardiography shows increased thickness of the left ventricular walls. An ECG recording is shown in the attached figure. Which of the following labeled components of the ECG most likely represents electrical depolarization of the thickened ventricle in this patient?

- ✓A. A
- B. B
- C. C
- D. D
- E. E

Rationale: This patient has an athlete's heart; a condition found in athletes with high intensity and prolonged training sessions for long periods of time. The left ventricle is enlarged as the physical examination findings show. The component of the ECG related to the electrical activity of this cavity is the QRS complex, which is labelled with letter A.

Attachment:



Picture2.png

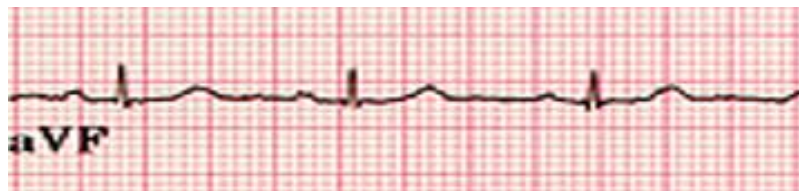
Question #: 13

A 61-year-old woman comes to the physician because of a 2-day history of chest discomfort. She has a 12-year history of migraines and uses a beta-blocker. Her pulse is 82/min, blood pressure is 138/82 mmHg, and respirations are 12/min. Physical examination shows no abnormalities. An ECG is shown in the attached figure. Which of the following lead placements most likely corresponds to the recording from the lead shown in the attached figure?

- ✓A. Positive electrode located in the left leg
- B. Positive electrode located in the left shoulder
- C. Positive electrode located in the right shoulder
- D. Negative electrode located in the left leg
- E. Negative electrode located in the left shoulder

Rationale: The lead aVF is represented in the image. This is a unipolar limb lead, with the positive electrode in the left leg. There is no negative electrode in this type of lead because it is unipolar.

Attachment:



Picture6.png

Question #: 14

A 20-year-old man comes to the physician because of a 1-week history of fever, persistent cough, chest pain, and shortness of breath. He is diagnosed with bronchopneumonia and treated with a 1-week course of antibiotics. An ECG recorded in leads I and II are shown in the attached figure. Which of the following is the most likely value of the mean electrical axis of this patient's heart?

- A. $+0^\circ$
- B. -30°
- ✓C. $+30^\circ$

D. $+60^\circ$

E. $+90^\circ$

Rationale: You will have to use leads 1 and 2 to calculate the MEA. Since leads 1 and 2 are both upright, you can quickly ascertain that there is normal axis deviation. You will have to graph out the MEA. Since the lead amplitudes are almost equal and thus the MEA is midway between 0 and 60 degree, that is 30 degrees.

Attachment:



ECG4.jpg

Question #: 15

A 45-year-old man is brought to the emergency department because of a 2-hour history of palpitations. He says that he is feeling "flutters" in his chest. He has a 10-year history of poorly controlled high blood pressure. His pulse is 104/min and has an irregular rhythm, blood pressure is 146/86 mmHg, and respirations are 18/min. Physical examination shows a loud S2 in the aortic area. An ECG shows atrial flutter. The net QRS amplitudes measured in the leads I and aVF show:

Lead I = 2 mV and

Lead aVF = 3 mV.

Which of the following leads is most likely to show the greatest net positive QRS amplitude?

- A. Lead I
- ✓B. Lead II
- C. Lead III
- D. Lead aVR
- E. Lead aVL

Rationale: This is essentially a mean electrical axis question. You have to determine which EKG lead closest to the MEA. Make sure that you graph the amplitudes in the direction of the leads recording direction.

Question #: 16

A 65-year-old man is brought to the emergency department because of a sudden onset of chest pain and shortness of breath. He has hypertension and type 2 diabetes mellitus. ECG reports show ST-elevation in anterolateral leads. Which of the following physiological changes is most likely present in this patient?

- A. Decreased systemic vascular resistance
- B. Increased cardiac contractility
- C. Decreased pulmonary capillary wedge pressure
- ✓D. Increased end-diastolic LV pressure
- E. Decreased pulmonary artery systolic pressure

Rationale: Decreased cardiac contractility will lead to increased end-diastolic LV pressure, which is transmitted backwards to increase pulmonary capillary wedge pressure

Question #: 17

A 68-year-old woman is brought to the emergency department because of a 2-hour history of shortness of breath and palpitations. She has no family history of heart disease. Her pulse ranges from 86-125/min, blood pressure is 142/76 mmHg, and respirations are 18/min. Physical examination shows an irregularly irregular pulse. An ECG shows atrial fibrillation. She is administered a Class III antiarrhythmic drug that blocks the delayed rectifier potassium channels in the cardiac myocytes which slow the process of repolarization. Which of the following ECG changes is most likely to be caused by this drug?

- A. Prolonged PR interval
- B. Widened QRS complex
- C. Irregular RR interval
- ✓D. Prolonged QT interval
- E. Decreased ST segment

Rationale: The drug reduces atrial arrhythmia by elongating the duration of repolarization of the action potentials and thus elongating the QT interval as well as increasing the effective refractory period.

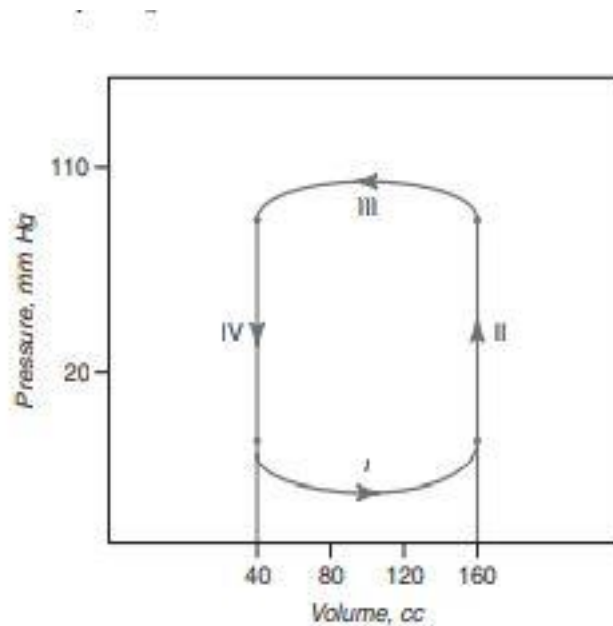
Question #: 18

A 54-year-old woman comes to the physician because of a 3-week history of persistent general malaise and fever. The symptoms developed slowly over the weeks following her root canal treatment 3 months ago. Her temperature is 38.3°C (101°F). On cardiac examination, a click and a systolic murmur are auscultated in the mitral area. She tells the physician that the click is due to a mechanical valve replacement done 4 years ago due to rheumatic fever as a child. The phases of a cardiac cycle are shown in the attached graph. Which of the following is the most likely phase of the cardiac cycle during which the murmur is heard in this patient?

- A. Phase I
- B. Phase II
- ✓C. Phase III
- D. Phase IV
- E. Not seen on this graph

Rationale: Phase III this is systolic ejection, when you would hear the systolic murmur of mitral regurgitation best

Attachment:



Reproduced, with permission, from USMLERx.com.

cardiac cycle.JPG

Question #: 19

A 76-year-old woman comes to the physician for a pre-operative evaluation for knee surgery. Her blood pressure is 200/120 mm Hg. While in the hospital room, she sings in order to relax, and her blood

pressure drops to 180/90 mm Hg. Using the attached image, which of the following vascular changes is most likely responsible for decreasing the patient's blood pressure?

- A. A
- B. B
- C. C
- ✓D. D
- E. E

Rationale: Singing impacts the cardiovascular system positively by improving the heart rate variability. Singing also follows the same principle of increased thoracic pressures which increases venous return. Relaxation of the peripheral muscles results in reduced systemic vascular resistance.

Attachment:

	Total peripheral resistance	Venous return
A	Increased	Increased
B	Decreased	Decreased
C	Increased	Decreased
D	Decreased	Increased
E	Increased	No change

CPRHemo.jpg

Question #: 20

An 18-year-old man comes to the physician for a routine examination. He has a family history of heart disease. He is very athletic and has been training for running competitions for several years. An ECG shows left axis deviation. An echocardiogram shows bradycardia, low but normal ejection fraction, and a dilated left ventricle with a thickened wall. There is no evidence of hypertension or valve disease. Which of the following hemodynamic changes is most likely in this patient?

- A. Increased systolic pressure
- B. Increased transmural pressure
- ✓C. Increased wall tension
- D. Increased diastolic pressure
- E. Decreased radius

Rationale: Dilated cardiomyopathy causes impaired mechanical function. A dilated ventricle has to generate increased wall tension to produce the same intraventricular pressure. Wall tension can be thought of as the tension generated by myocytes that results in a given intraventricular pressure at a particular ventricular radius.

Question #: 21

A 40-year-old woman volunteers to participate in a study examining the physiological effects of breath holding. While sitting, she holds her breath for 2 minutes (prolonged phase 2 Valsalva). During this apnea, her pulse increases from 60 to 76/min, mean arterial pressure decreases, and stroke volume decreases from 100 ml to 50 ml. The reduction in the patient's stroke volume during the apnea most likely results from a decrease in which of the following parameters?

- A. Venous compression
- B. Intrachamber pressures
- C. Cardiac compression
- ✓D. Cardiac filling

Rationale: Initially during a Valsalva, intrathoracic (intrapleural) pressure becomes very positive due to compression of the thoracic organs by the contracting rib cage. This increased external pressure on the heart and thoracic blood vessels compresses the vessels and cardiac chambers by decreasing the transmural pressure across their walls. Venous compression, and the accompanying large increase in right atrial pressure, impedes venous return into the thorax. This reduced venous return, and along with compression of the cardiac chambers, reduces cardiac filling and preload despite a large increase in intrachamber pressures. Reduced filling and preload leads to a fall in cardiac output by the Frank-Starling mechanism.

Question #: 22

A 50-year-old woman comes to the physician because of a 2-month history of hypertension. She has a strong family history of coronary artery disease. Her blood pressure has ranged from 150/95mmHg to 130/85mmHg during the previous three visits within the past two months. Her blood pressure today is 140/90mmHg. Physical examination shows no other abnormalities. Which of the following values best approximates her mean arterial pressure today?

- A. 119 mm Hg
- B. 113 mm Hg
- ✓C. 107 mm Hg
- D. 101 mm Hg
- E. 95 mm Hg

Rationale: The correct answer is 107 mm Hg. A rough approximation of $MAP = DP + (SBP - DPP)/3$. Given the values from the vignette, we would get $MAP = 90 + (140-90)/3 = 107$

Question #: 23

During an experiment, four different vessels of the same length are sampled. An indicator dye is mixed with blood and injected through the vessel. A decrease in which of the following would most likely be expected to increase the blood flow in the vessel?

- A. Plasma sodium concentration
- B. Plasma oncotic pressure
- C. Radius of the vessel
- D. Pressure gradient across the vessel
- ✓E. Viscosity of the blood

Rationale: The correct answer is 'Viscosity of the blood'. Flow is directly proportional to the pressure gradient across the vessel. Flow is also directly proportional to the radius of the vessel to the fourth power. Flow is inversely proportional to the viscosity of the blood. Therefore, decreased viscosity is equal to an increased blood flow.

Question #: 24

48-year-old obese man comes to the physician because of a 10-month history of lower leg pain that occurs after walking long distances. It is relieved with rest. He has a 10-year history of hyperlipidemia. His blood pressure is 185/85 mm Hg, pulse is 83/min, and his respiratory rate is 18/min. Doppler studies of the lower limbs show increased arterial pressure in the major feeding arteries. Examination of a biopsy specimen of these arteries shows arteriosclerosis. Which of the following physiological changes in the affected arteries is most likely the cause of elevated pressure in this patient?

- A. increased laminar flow
- B. increased elasticity
- C. increased radius
- ✓D. decreased compliance
- E. decreased tension

Rationale: Blood pressure has been shown to be a strong predictor of cardiovascular risk—increased systolic blood pressure (SBP) reflecting stiffening of the arterial walls and changes in vascular structure, and increased pulse pressure (PP) reflecting stiffening of conduit vessels. Arterial stiffness may now be measured as a marker of arterial health, and in combination with cardiac output can be used to calculate total arterial compliance. this patient has arteriosclerosis that affects his vasculature. We expect this will reduce the radius, reduce the elasticity, increase turbulent flow, decrease compliance, and increase wall tension.

Question #: 25

A study was conducted to determine the hemodynamic effect of a drug on chronic heart failure. The participants of the study included patients with decreased contractile force and mechanically impaired left ventricle filling. The results of the study showed that the drug had a positive inotropic effect. Which of the following drugs was most likely used in the study?

- A. Beta receptor antagonist
- ✓B. Cardiac glycoside
- C. Diuretic
- D. ACE inhibitor
- E. Angiotensin II receptor antagonist

Rationale: Cardiac glycosides is the correct answer, digoxin is an example of this type of drug. It is used to treat heart failure and to control ventricular rate in atrial fibrillation. It belongs to a class of medications called cardiac glycosides. It has positive inotropic and negative chronotropic activity.

Question #: 26

A 77-year-old man comes to the physician because of a 6-month history of swelling of his legs and feet. He has a 40-year history of alcoholism and a 5-year history of hepatic disease. Physical examination shows ascites and a 2+ edema of the lower extremities. A decrease in which of the following most likely promotes edema formation in this patient?

- A. Capillary hydrostatic pressure
- B. Filtration coefficient
- C. Interstitial colloid osmotic pressure
- D. Interstitial fluid hydrostatic pressure
- ✓E. Plasma colloid oncotic pressure

Rationale: Oncotic pressure of the plasma is primarily maintained by albumin. Reduced concentration of albumin in plasma (hypoalbuminemia) may result from: Decreased protein synthesis: Most plasma

proteins are synthesized in the liver.

Question #: 27

A 69-year-old man comes to the physician because of a 2-day history of black, tarry stools. He has a 30-year history of alcoholism and liver disease. His pulse is 82/min, blood pressure is 142/86 mmHg, and respirations are 16/min. Physical examination shows abdominal distension. Abdominal ultrasound shows the presence of ascites. If the values that determine the capillary fluid exchange are as follows:

Hydrostatic capillary pressure = 21 mmHg

Capillary oncotic pressure = 6 mmHg

Tissue interstitial pressure = 10 mmHg

Tissue oncotic pressure = 6 mmHg

Which of the following values best approximates the net driving force?

- A. - 10 mmHg
- ✓B. + 11 mmHg
- C. + 14 mmHg
- D. + 16 mmHg
- E. + 22 mmHg

Rationale: This question obviously requires use of the Starlings equation to determine fluid movement in the microcirculation. They are asking for the filtration pressures and not flux, and no reflection coefficient are given (to adjust for capillary permeability). So, we just plug in the values: $NFP = (\text{hydrostatic } P (\text{capillary} - \text{interstitial})) - (\text{oncotic pressures hydrostatic } P (\text{capillary} - \text{interstitial}))$. $NFP = (21 - 10) - (6 - 6) = 11 \text{ mmHg}$

The symptoms of portal hypertension include esophageal varices (enlarged vessels) that can burst and have the patient spit up red blood. The tarry stools indicate blood that has bled through the digestive system (swallowed blood).

Question #: 28

A 64-year-old man comes to the physician because of a 2-day history of cough and shortness of breath. He has a 20-year history of high blood pressure. His pulse is 108/min, blood pressure is 168/92 mmHg, and respirations are 16/min. Auscultation of the lungs shows crackles in both lung fields. An x-ray of the chest shows bronchopneumonia. The patient is admitted to the ward and intravenous infusion with isotonic fluid is started. Which of the following changes in the venous return curve is most likely a result of the intravenous infusion?

- A. A

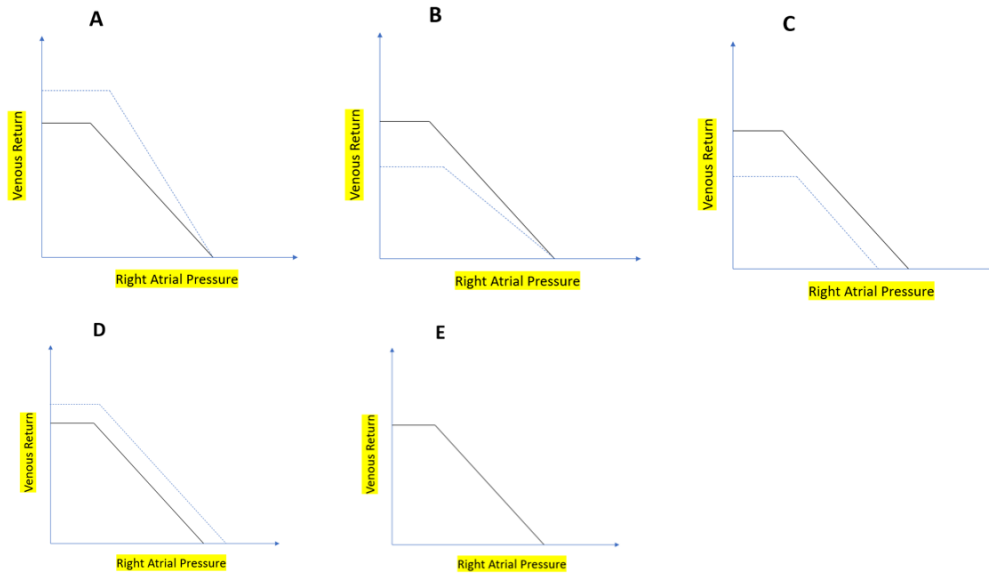
- B. B
- C. C
- ✓D. D
- E. E

Rationale: The patient has symptoms of congestive heart failure. In heart failure there is fluid retention leading to an increase in total body water causing an increase in MSFP and an increase in VR.

Accordingly, the correct answer is D.

Attachment:

Solid line: normal
Dotted line: patient



Capture.PNG

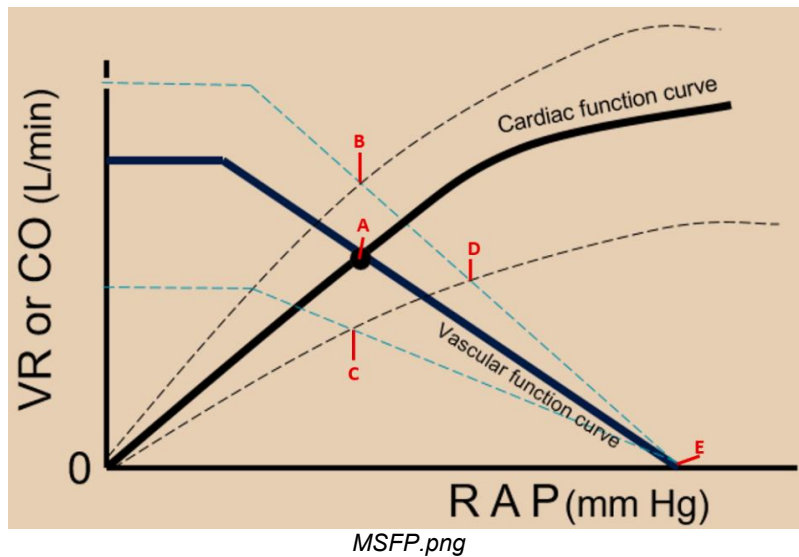
Question #: 29

A 45-year-old man comes to the emergency room because of chest pains for the past 3-weeks which worsens on deep inspiration. His pulse is 88/min, his respirations are 14/min, and his blood pressure is 120/80 mmHg. His Guyton cross plot is shown in the attached image. Which of the following labeled points best represents his mean systemic filling pressure?

- A. A
- B. B
- C. C
- D. D
- ✓E. E

Rationale: The point E on the graph represents the mean systemic filling pressure. This pressure creates a pressure difference that allows blood flow into the right atrium.

Attachment:



Question #: 30

A 71-year-old man comes to the physician because of a 3-day history of pain and swelling in his left leg. He underwent an operation for intestinal occlusion 5 weeks ago. Physical examination shows marked swelling of the leg with dilated superficial veins. Doppler ultrasound shows an extensive left leg deep venous thrombosis, with complete occlusion. Anticoagulation and thrombolytic treatments are commenced. Following treatment, which of the following mechanisms most likely controls local blood flow to the skeletal muscle distal to the occlusion?

- A. Active hyperemia
- ✓B. Reactive hyperemia
- C. Myogenic response
- D. Prostacyclin mediated vasodilation
- E. Sympathetic mediated vascular tone

Rationale: Reactive hyperemia is the transient increase in organ blood flow that occurs following a brief period of ischemia (e.g., arterial occlusion). It occurs following the removal of a tourniquet, unclamping an artery during surgery, or restoring flow to a coronary artery after recanalization (reopening a closed artery using an angioplasty balloon or clot dissolving drug).